A single institution experience of the incidence of extracranial metastasis in glioma.

PMID: 22595352 | Indexed in PubMed

Glioblastoma multiforme (GBM) is the most common and aggressive subtype of all gliomas. The prognosis is poor but despite the aggressiveness of the tumour, extracranial metastasis of glioma is rare. Most documented cases of extracranial metastases of GBM involve leptomeningeal spread to the spine. In this clinical study we aim to review the incidence and location of extracranial metastasis of glioma from the Brain Tumor Database of the National Neuroscience Institute of Singapore, between September 2004 to October 2009. Four of 148 patients (2.7%) were identified, one of whom had pleomorphic xanthoastrocytoma (PXA) with scalp and spinal metastasis, suggestive of haematogenous rather than cerebrospinal fluid spread that has been described elsewhere. To our knowledge, there has been no published report of PXA with scalp metastasis or vertebral metastasis.

Bone metastasis from glioblastoma: a systematic review.

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Glioblastoma (GBM) is a devastating disease with poor overall survival. Despite the common occurrence of GBM among primary brain tumors, metastatic disease is rare. Our goal was to perform a systematic literature review on GBM with osseous metastases and understand the rate of metastasis to the vertebral column as compared to the remainder of the skeleton, and how this histology would fit into our current paradigm of treatment for bone metastases. A Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA)-compliant literature search was performed using the PubMed database from 1952 to 2021. Search terms included "GBM", "glioblastoma", "high-grade glioma", "bone metastasis", and "bone metastases". Of 659 studies initially identified, 67 articles were included in the current review. From these 67 articles, a total of 92 distinct patient case presentations of metastatic glioblastoma to bone were identified. Of these cases, 58 (63%) involved the vertebral column while the remainder involved lesions within the skull, sternum, rib cage, and appendicular skeleton. Metastatic dissemination of GBM to bone occurs. While the true incidence is unknown, workup for metastatic disease, especially involving the spinal column, is warranted in symptomatic patients. Lastly, management of patients with GBM vertebral column metastases can follow the International Spine Oncology Consortium two-step multidisciplinary algorithm for the management of spinal metastases.

Systemic metastasis in glioblastoma may represent the emergence of neoplastic subclones.

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Glioblastomas only rarely metastasize to sites outside the central nervous system, for reasons that are poorly understood. We report the clinicopathological and molecular genetic findings in 6 patients with metastatic glioblastoma. Four patients were under the age of 32 and all but 1 patient died within 2 yr of diagnosis. The number of metastases ranged from 1 to 3. At the time of death, 3 patients had apparent tumor control at their primary site. We evaluated DNA from both primary and metastatic glioblastomas for genetic alterations commonly found in glioblastomas: TP53 mutations, CDKN2A/p16 deletions, EGFR amplification, and allelic loss of chromosomes 1p, 10q and 19q. Four of 6 cases had TP53 mutations and only single cases had EGFR amplification, CDKN2A/p16 deletions, or allelic loss of 1p, 10q and 19q; 2 cases had no detectable genetic alterations. In 2 cases, the primary and metastatic tumors had identical genotypes. Remarkably, however, 2 cases had different TP53 alterations in the primary and metastatic lesions, or among the metastatic tumors, which suggests that some metastatic deposits may represent emergence of subclones that were not necessarily dominant in the primary tumor. The present observations and a review of the recent literature demonstrate that metastatic glioblastomas tend to occur in younger adults who do not follow long clinical courses, and may be characterized by TP53 mutations and differential clonal selection.

Variant CD44 adhesion molecules are expressed in human brain metastases but not in glioblastomas.

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Although human glioblastomas are highly invasive tumors intracerebrally, only rarely do they metastasize outside the central nervous system. In contrast, the brain is a major target for metastatic spread of many systemic tumors. Recently, it was demonstrated that expression of splice variants of CD44 (CD44v), but not standard CD44 (CD44s), was sufficient to confer metastatic potential to low- or nonmetastatic rat tumor cells. Because CD44 is expressed in brain tumors, we examined whether differential expression of CD44 isoforms was correlated with the metastatic behavior of these tumors. We compared CD44s and CD44v expression in 17 human glioblastomas, 18 glioma cell lines, and metastases of 15 other tumors to the brain by reverse transcription/polymerase chain reaction, Northern blotting, and immunocytochemistry. These experiments showed that 0 of 17 glioblastomas and 0 of 18 glioma cell lines expressed CD44v as compared to 12 of 15 brain metastases. These data show a correlation between CD44v expression and the metastatic ability of the tumors analyzed ($P < 0.01$). This suggests (a) that the biological significance of the lack of CD44v expression in human glioblastomas warrants further examination with regard to their inability to metastasize extraneurally and (b) that CD44v expression may play a role in the intracerebral spread of about 80% [corrected] of the brain metastases. Therefore, CD44v expression should be further considered as a potential marker for differential diagnosis and prognosis of patients with brain metastases.

Secondary glioblastoma metastasis outside the central nervous system in a young HIV-infected patient.

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Glioblastoma is the most common adult primary brain tumor that occurs in the central nervous system and is characterized by rapid growth and diffuse invasiveness with respect to the adjacent brain parenchyma, which renders surgical resection inefficient. Although it is a highly infiltrative tumor, it is rarely disseminated beyond the central nervous system, wherein extracranial metastasis is a unique but rare manifestation of this kind of tumor. It is very common for acquired immunodeficiency syndrome (AIDS) patients to be infected with the human immunodeficiency virus (HIV), which suggests that a possible association between HIV infection and tumor development exists. In this paper, we present a new case of a young patient's HIV-associated glioblastoma, with glioblastoma metastasis within the T9 vertebral body and lymph nodes in the anterior neck tissue. Initially, the patient was diagnosed with a grade III plastic astrocytoma. The patient lived a normal life for a year while being treated with temozolomide, radiotherapy, and highly active antiretroviral therapy. However, the tumor quickly evolved into a glioblastoma. We believe that the drastic progression of the tumor from a grade III anaplastic astrocytoma to a metastatic glioblastoma is due to the HIV infection that the patient had acquired, which contributed to a weakened immune system, thus accelerating progression of the cancer.

Extracranial metastases of glioblastoma in a child: case report and review of the literature.

PMID: 17356401 | Indexed in PubMed

Glioblastoma (GBM) is the most common adult malignant brain tumor but is notably less common in children. Primary brain tumors rarely metastasize outside the central nervous system and when metastases occur, it is often in patients with diversionary shunting of the cerebrospinal fluid. This report details the case of a 13(1/2)-year-old boy who was diagnosed with GBM. He survived 10 months after diagnosis. At autopsy, the tumor was found to extensively infiltrate the leptomeninges as well as the cranial skin and soft tissue. Further examination disclosed multiple liver and lung metastatic GBM nodules. This pattern of spread is very uncharacteristic of gliomas and emphasizes the importance of adequate metastatic evaluation.

Biologic characterization of a secondary glioblastoma with extracranial progression and systemic metastasis.

PMID: 12626129 | Indexed in PubMed

Glioblastomas rarely metastasize outside the CNS. We biologically characterized a case of secondary glioblastoma associated with extracranial progression and distant metastasis. A 42-year-old male patient was subjected to craniotomy for a left temporal tumor (astrocytoma grade II) and subsequently underwent another 3 craniotomies due to tumor recurrences. At the third craniotomy, extracranial progression was noted, and the tumor was classified as a glioblastoma. In order to pinpoint the genes expressed differentially in the intracranial primary tumor and the metastatic tumors, we used cDNA microarray. The patterns of gene expression in these 2 samples were highly similar, suggesting that the mechanism of metastasis was direct infiltration of tumor cells into extracranial blood vessels. Insulin-like growth factor binding protein-2 was overexpressed in both primary and metastatic tumors. Immunohistochemical studies of DNA-dependent protein kinase, which participates in the repair of DNA, was strongly positive in the samples obtained at the first and second operations, but the positive rates were markedly reduced in the specimens obtained at the third and fourth operations. These results suggest that insulin-like growth factor binding protein-2 and deficiency of DNA-dependent protein kinase proteins promoted tumor progression in the present case.

Report of GBM metastasis to the parotid gland.

PMID: 16132510 | Indexed in PubMed

Glioblastoma Multiforme frequently metastasises from their original location by for example infiltration along white matter tracts [1]. GBM metastasis outside the central nervous system is distinctly rare though there are previous reports of spread to various organs [2-5]. We add an unusual case of a patient with aggressive cerebral GBM metastasis to the parotid gland and the lungs.

Infratemporal and intraorbital metastasis of the glioblastoma multiforme. A case report.

PMID: 15213638 | Indexed in PubMed

Primary brain tumors rarely metastasize outside of the central nervous system. A case of 37-year-old woman with a temporal lobe glial tumor which shows infratemporal and orbital invasion is presented. She had undergone a cranial operation because of temporal lobe epilepsy 8 years ago. Defects on the dura and bone flap were thought to be the mechanisms of the infratemporal fossa invasion. The case illustrates extra-axial metastasis of glioblastoma multiforme, an exceptional finding.

Unusual Case of Transdural Extension of Glioblastoma Multiforme in the Maxillofacial Region and a Review of the Literature.

PMID: 31348875 | Indexed in PubMed

Glioblastoma multiforme is the most common primary brain tumor. It is locally aggressive but rarely spreads outside the central nervous system. We present an unusual case of a 57-year-old woman who had presented 1 year after surgical resection and adjuvant therapy, with evidence of recurrent tumor invading through the skull base into the orbital apex, masticator, and pterygoid space. We have also reviewed all available case reports on local invasion of glioblastoma that occurred in absence of treatment and recurrence that developed away from the initial surgical location.
